C PROGRAMMING

Lesson 8 — Switch Statements

Based on: Bro Code C Tutorial Series

1. What is a Switch Statement?

A switch statement is a cleaner alternative to a long chain of if/else if statements. Instead of checking the same variable over and over, switch evaluates the expression once and jumps directly to the matching case.

Use switch when you are checking one variable against many fixed values — like days, menu options, or grades.

> **Limitation:** Switch only works with integers and characters. It does NOT work with floats, doubles, or strings.

2. Basic Syntax

```
switch (expression) {
    case value1:
        // code to run
        break;
    case value2:
        // code to run
        break;
    default:
        // runs if no case matches
}
```

- **expression —** the variable being checked (int or char)
- **case value: —** the value to match against
- **break —** exits the switch after a match (very important!)
- **default —** runs if none of the cases match (like the final else)

3. Example — Days of the Week (Integer)

```
int day = 0;
printf("Enter a day number (1-7): ");
scanf("%d", &day);

switch (day) {
    case 1:
```

```
        printf("Monday\n");
        break;
    case 2:
        printf("Tuesday\n");
        break;
    case 3:
        printf("Wednesday\n");
        break;
    case 4:
        printf("Thursday\n");
        break;
    case 5:
        printf("Friday\n");
        break;
    case 6:
        printf("Saturday\n");
        break;
    case 7:
        printf("Sunday\n");
        break;
    default:
        printf("Invalid! Enter a number between 1 and 7\n");
}
```

4. The Break Statement — Do NOT Forget This!

Without break, C keeps running the code in every case below the matching one. This is called fall-through and is almost always a bug.

```
// WITHOUT break -- fall-through bug!
switch (day) {
    case 1:
        printf("Monday\n");    // matches here...
    case 2:
        printf("Tuesday\n");   // ...but this also runs!
    case 3:
        printf("Wednesday\n"); // ...and this too!
}

// Input: 1
// Output: Monday
//         Tuesday
//         Wednesday   <-- not what we wanted!
```

> **Rule:** Always put a break at the end of every case unless you intentionally want fall-through (which is rare). Forgetting break is one of the most common C bugs.

5. Example — Days of the Week (Character)

Switch works with characters too. Use single quotes around character values in each case.

```
char c = '\0';
printf("Enter a day letter: ");
scanf(" %c", &c);

switch (c) {
    case 'M':
        printf("Monday\n");
        break;
    case 'T':
        printf("Tuesday\n");
        break;
    case 'W':
        printf("Wednesday\n");
        break;
    case 'R':
        printf("Thursday\n");   // R to avoid clash with Tuesday
        break;
    case 'F':
        printf("Friday\n");
        break;
    case 'S':
        printf("Saturday\n");
        break;
    case 'U':
        printf("Sunday\n");     // U to avoid clash with Saturday
        break;
    default:
        printf("Invalid character!\n");
}
```

6. Practice Problem

Practice: Simple Calculator Menu

Write a C program that shows the user a menu of operations, accepts their choice and two numbers, then performs the calculation.

Menu:

- 1 — Addition
- 2 — Subtraction
- 3 — Multiplication
- 4 — Division
- Any other number — Invalid choice

Expected Output (for choice 2, numbers 10 and 4)

```
--- Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division

Enter your choice: 2
Enter two numbers: 10 4

10 - 4 = 6
```